Software Engineering (IT632)
Dataflow Design Models

Winter 2026

Project Title :

CRAVIT - Smart Food Ordering and Delivery Platform

A web-based food ordering and delivery management system

Date of Submission: **15/03/2026**

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DAU, Winter 2026

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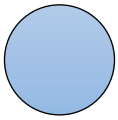
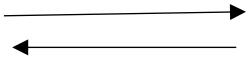
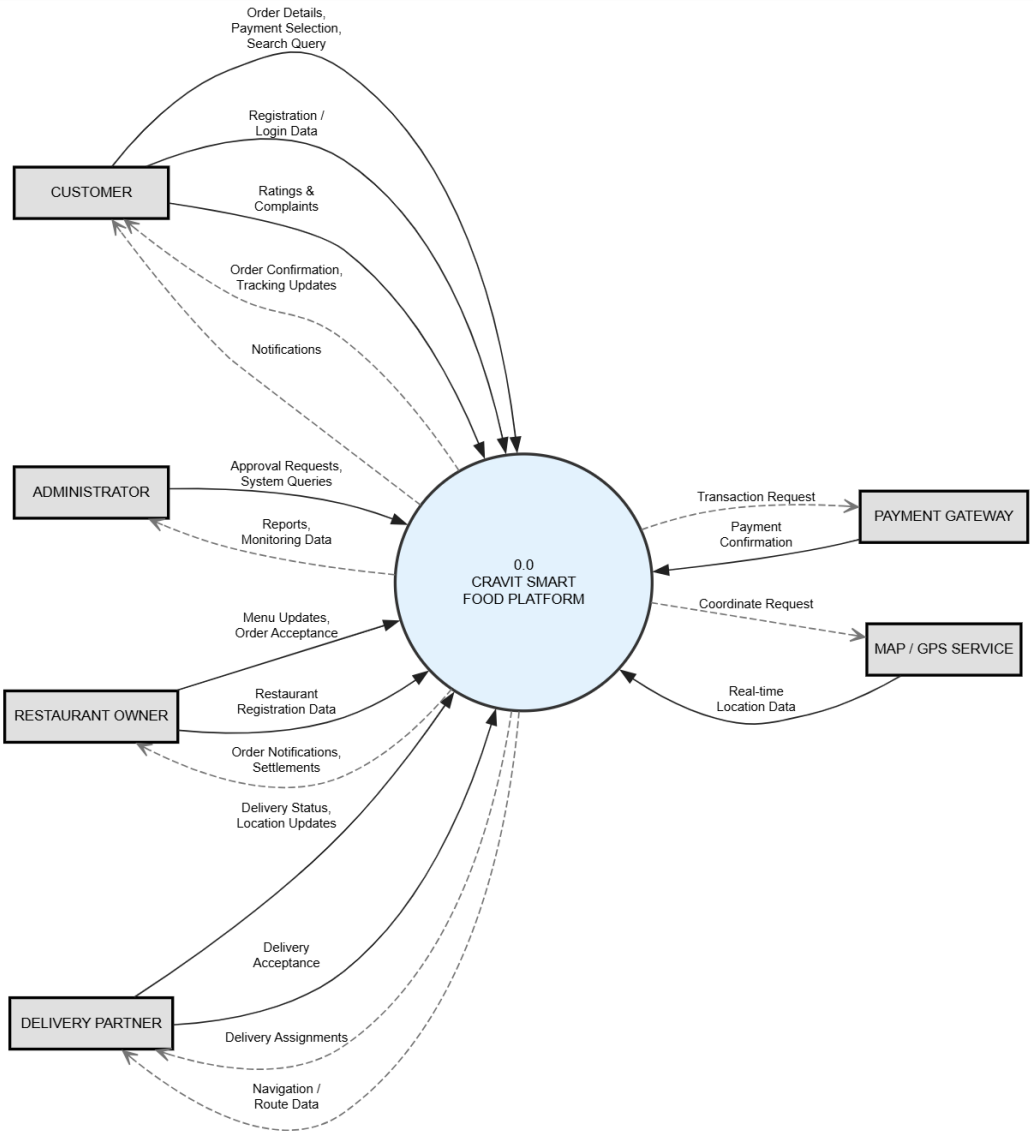
1.DFDs

The data flow in the Cravit Smart Food Ordering System represents how order, payment, and delivery information moves between customers, restaurants, delivery partners, and the system.

DFD Symbols :

Symbol	Name	Reperesentation
Rectangle	External Entity	
Circle	Process	
Two Parallel Boxes	Data Store	
Arrow Line	Data Flow	

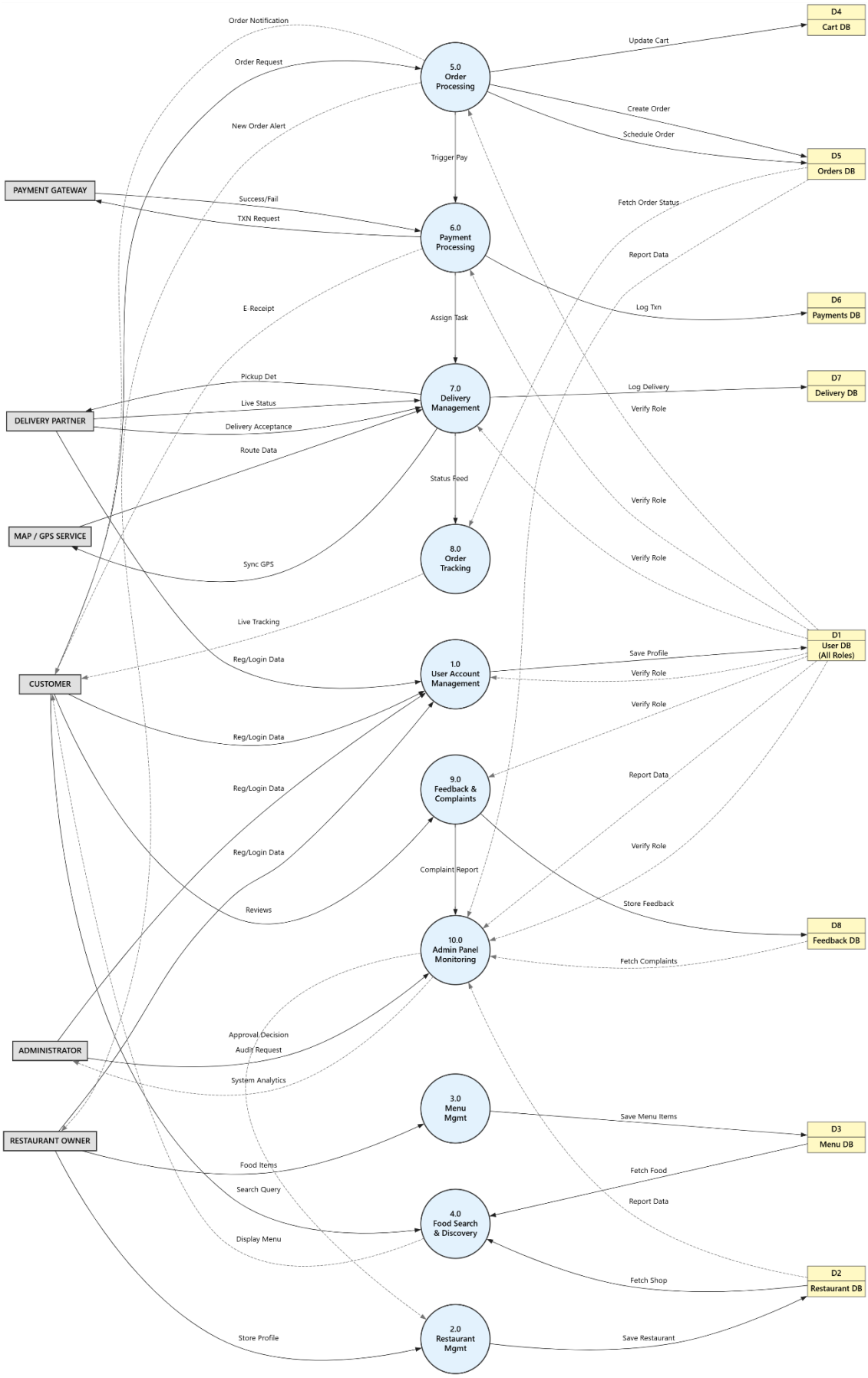
DFD Level 0(Context Level Diagram) :



Explanation:

- The Level 0 DFD (Context Diagram) represents the overall view of the Cravit Smart Food Ordering System.
- The entire system is shown as one main process (0.0 Cravit Smart Food Platform).
- It shows interaction between the system and external entities such as Customer, Administrator, Restaurant Owner, Delivery Partner, Payment Gateway, and Map/GPS Service.
- It illustrates how data such as orders, menu updates, payment information, and delivery updates flow between the system and these entities.
- This diagram provides a high-level understanding of the system without showing internal processes.

DFD Level 1 :


PAYMENT GATEWAY
DELIVERY PARTNER
MAP / GPS SERVICE
CUSTOMER
ADMINISTRATOR
RESTAURANT OWNER
5.0 Order Processing
6.0 Payment Processing
7.0 Delivery Management
8.0 Order Tracking
1.0 User Account Management
9.0 Feedback & Complaints
10.0 Admin Panel Monitoring
3.0 Menu Mgmt
4.0 Food Search & Discovery
2.0 Restaurant Mgmt
D4 Cart DB
D5 Orders DB
D6 Payments DB
D7 Delivery DB
D1 User DB (All Roles)
D8 Feedback DB
D3 Menu DB
D2 Restaurant DB
Order Notification
Order Request
New Order Alert
Update Cart
Create Order
Schedule Order
Trigger Pay
Success/Fail
TXN Request
Fetch Order Status
Report Data
Log Txn
E-Receipt
Assign Task
Pickup Det
Live Status
Delivery Acceptance
Route Data
Log Delivery
Verify Role
Status Feed
Sync GPS
Live Tracking
Reg/Login Data
Save Profile
Complaint Report
Reviews
Store Feedback
Fetch Complaints
Approval Decision
Audit Request
System Analytics
Save Menu Items
Food Items
Search Query
Fetch Food
Display Menu
Fetch Shop
Store Profile
Save Restaurant

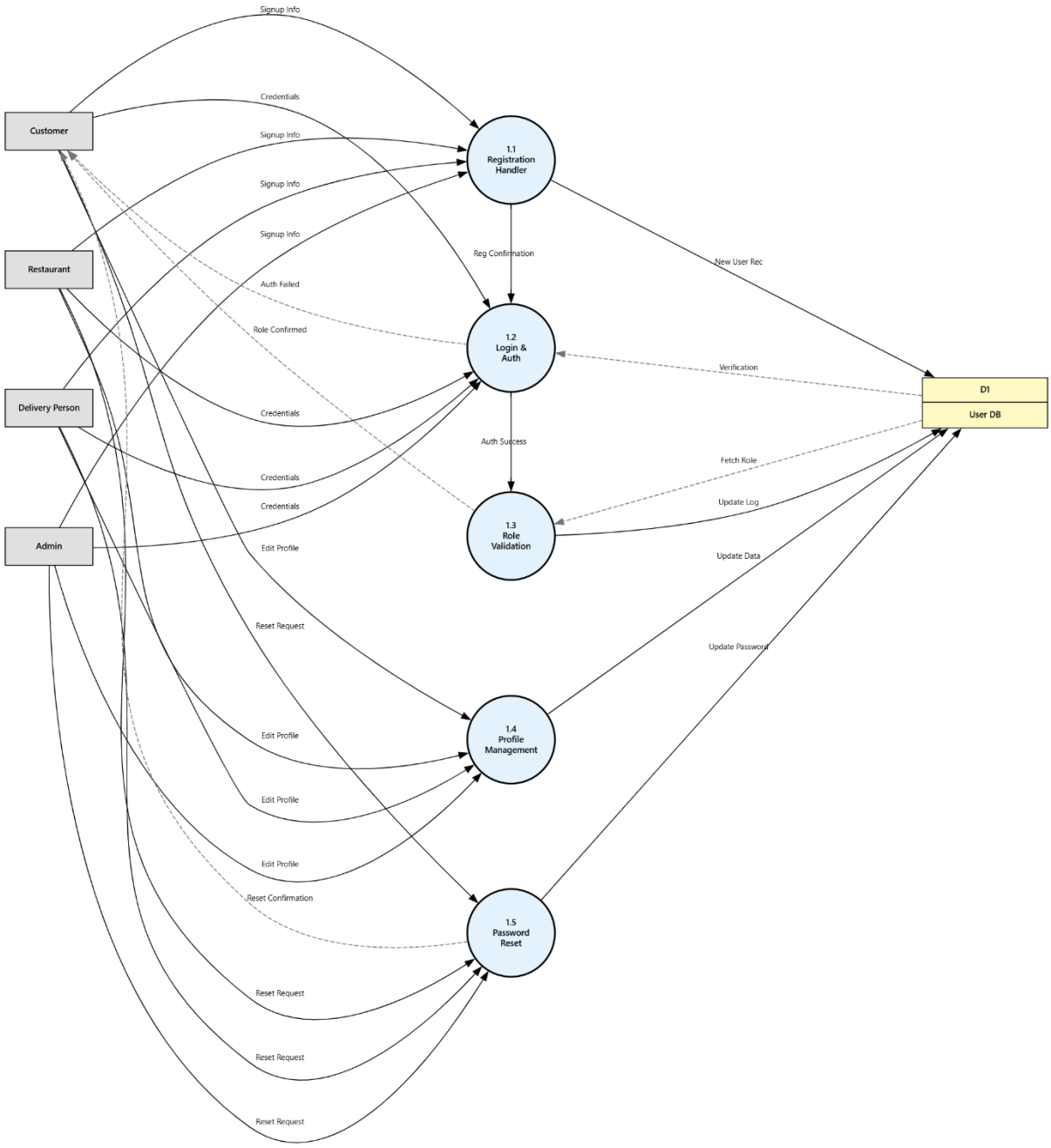
Explanation:

The Level 1 Data Flow Diagram provides a more detailed view of the internal processes of the Cravit Smart Food Ordering System.

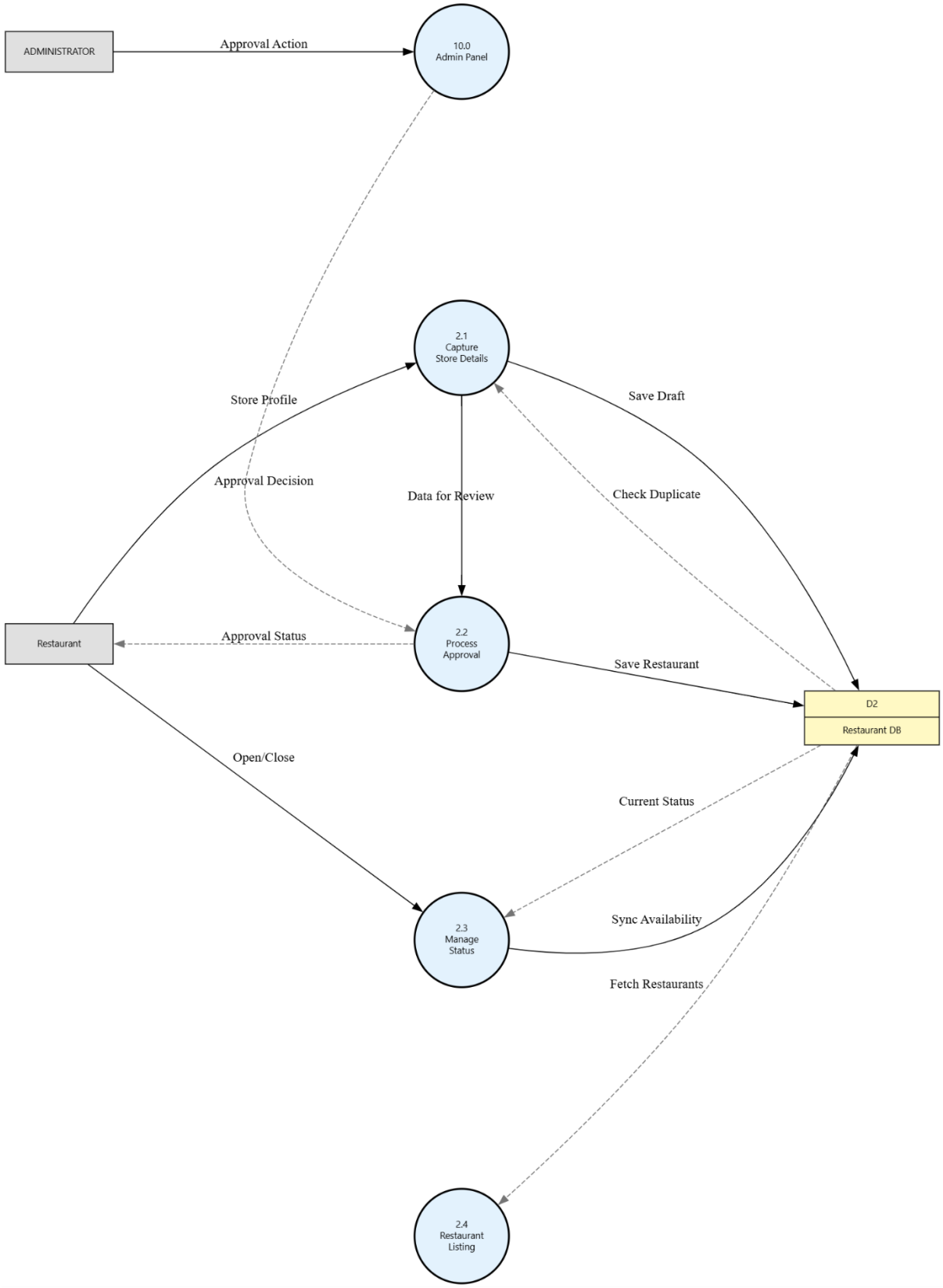
- It breaks the system into multiple functional processes such as User Account Management, Restaurant Management, Menu Management, Food Search, Order Processing, Payment Processing, Delivery Management, Order Tracking, Feedback & Complaints, and Admin Monitoring.
- Customers can search food, place orders, make payments, track deliveries, and give feedback through the system.
- Restaurant owners manage restaurant details and update menu items, while delivery partners receive delivery assignments and update delivery status.
- The system interacts with external services such as the Payment Gateway for transaction processing and Map/GPS Service for location and route data.
- Different data stores (databases) such as User DB, Restaurant DB, Menu DB, Orders DB, Payments DB, Delivery DB, and Feedback DB are used to store and retrieve system data.
- This level provides a detailed understanding of how data moves between processes, entities, and databases within the system.

DFD Level 2 :

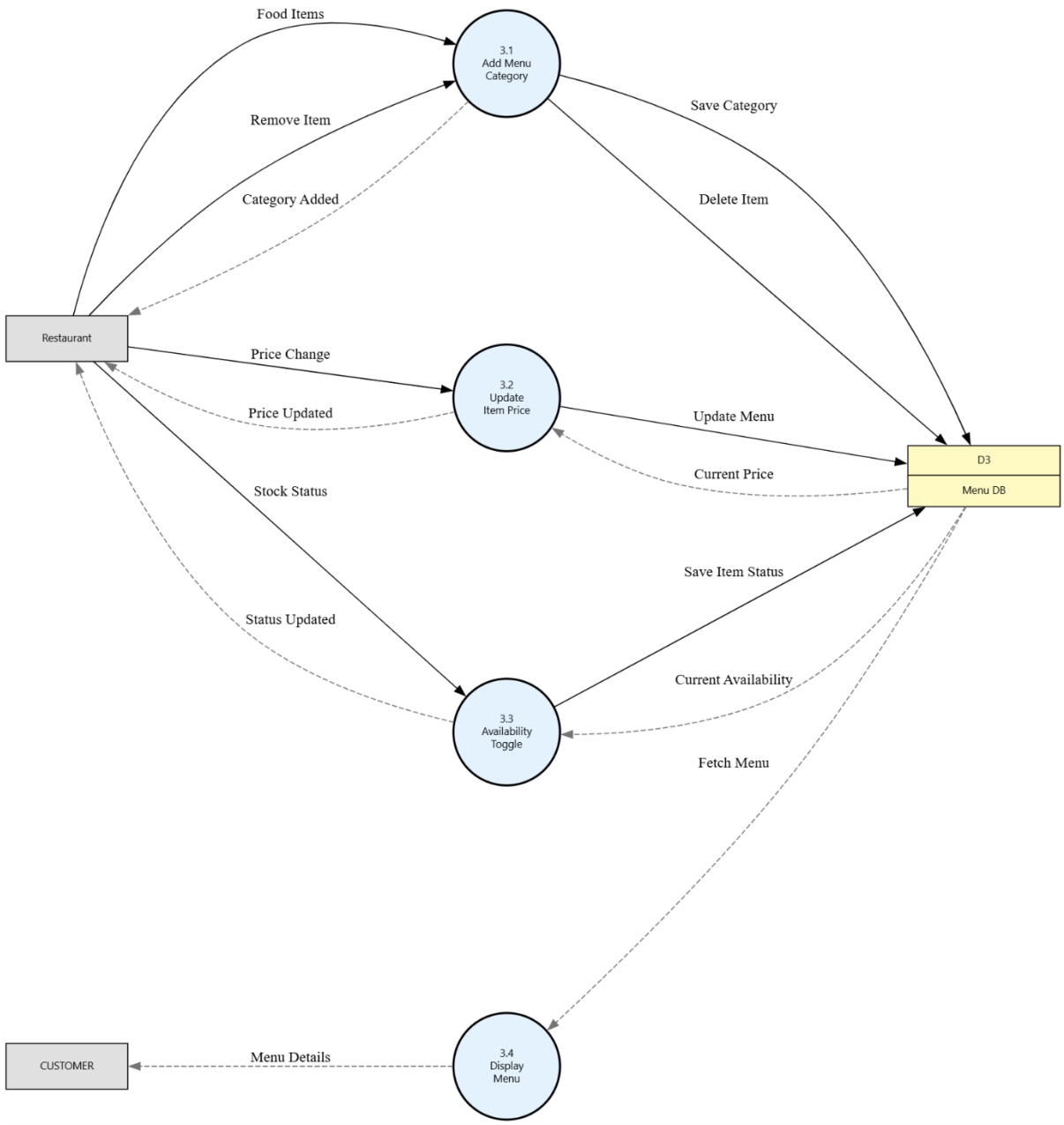
Process 1: User Account Management:



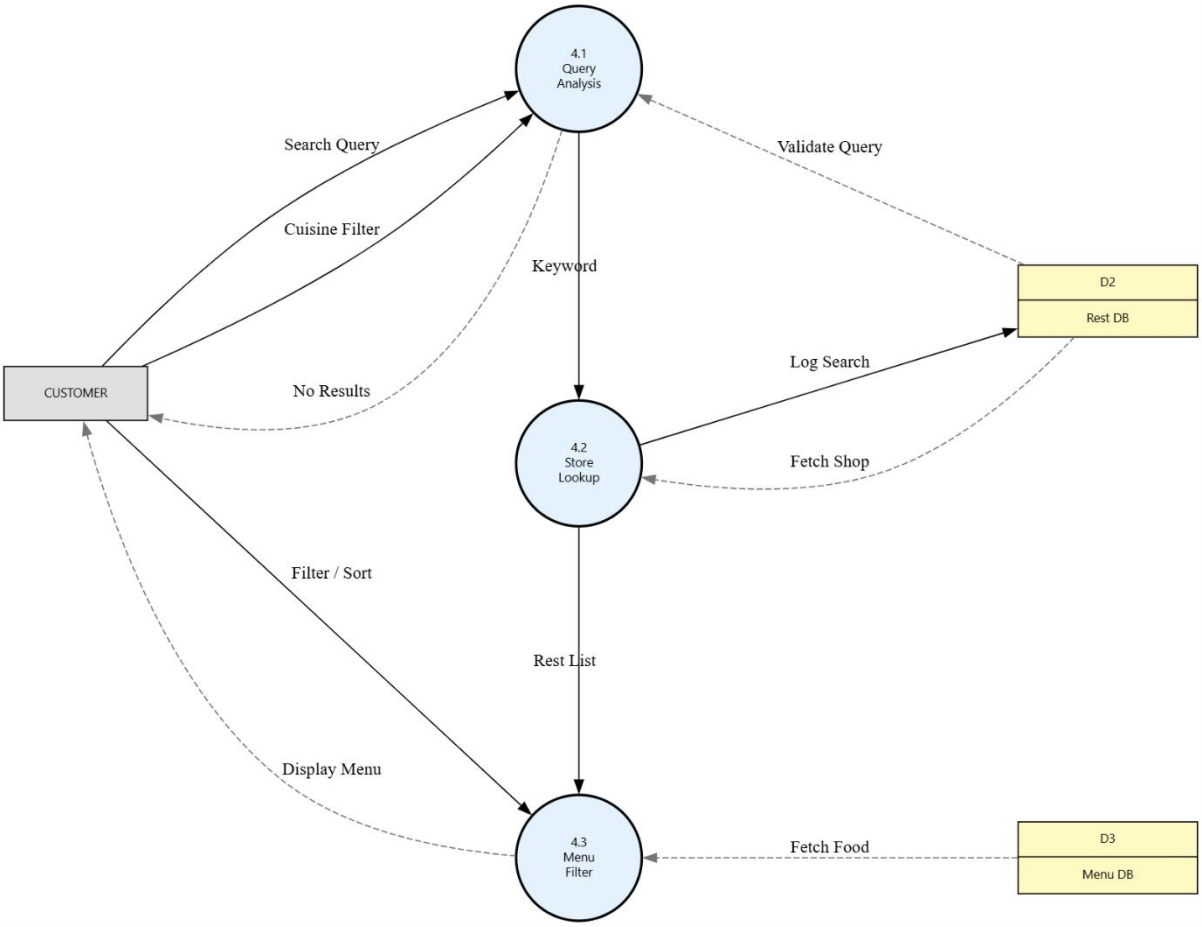
Process 2: Restaurant Management:



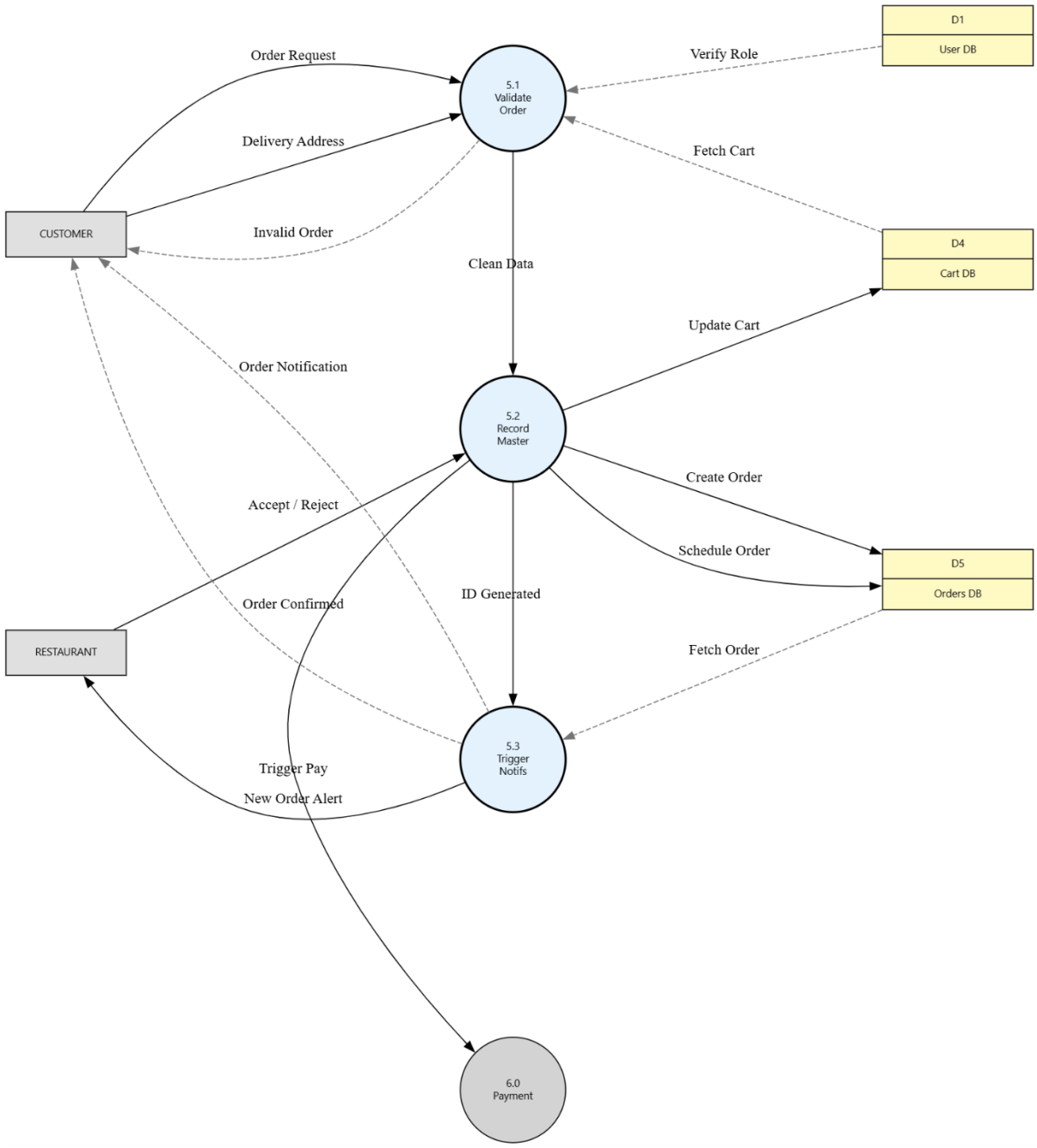
Process 3: Menu Management:



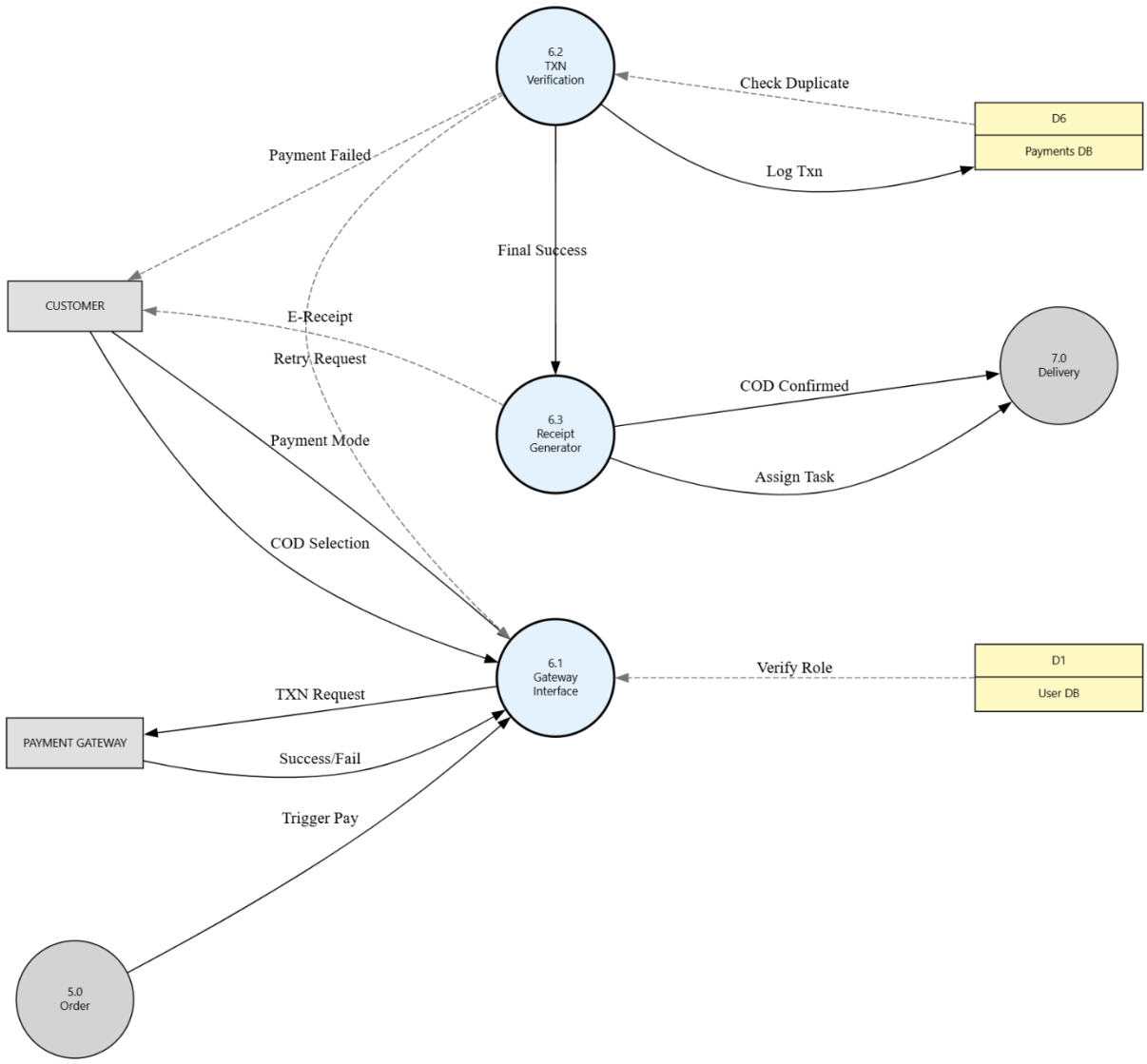
Process 4: Food Search and Discovery:



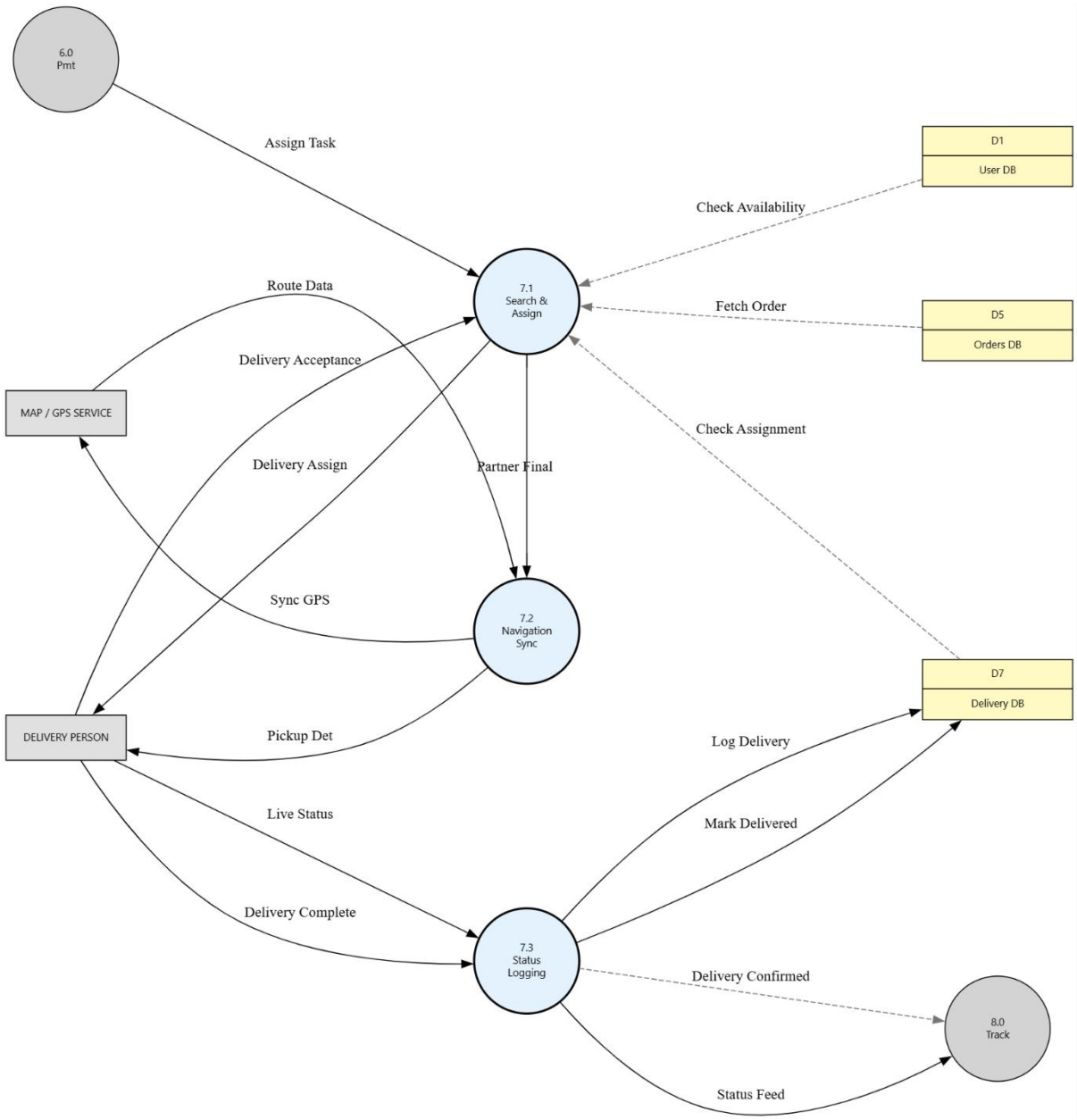
Process 5: Order Processing:



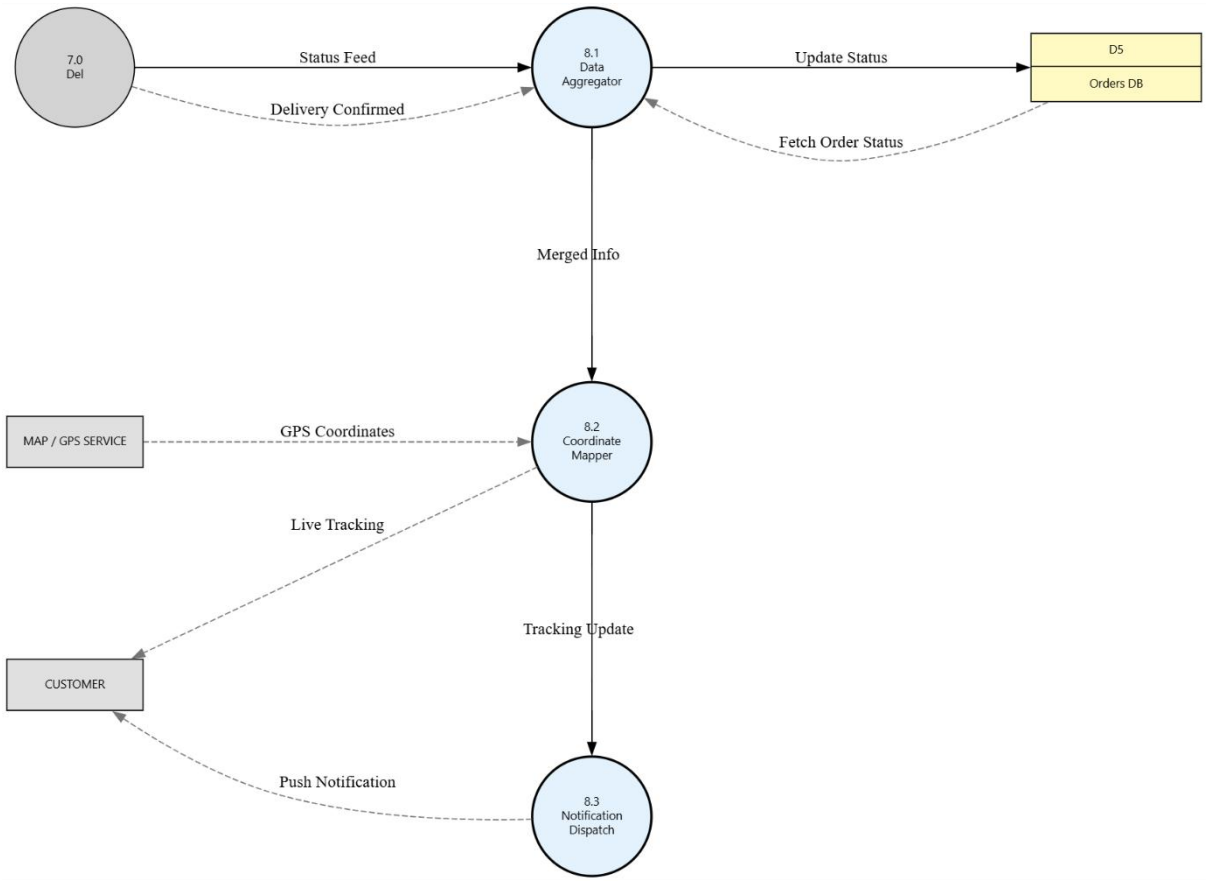
Process 6: Payment Processing:



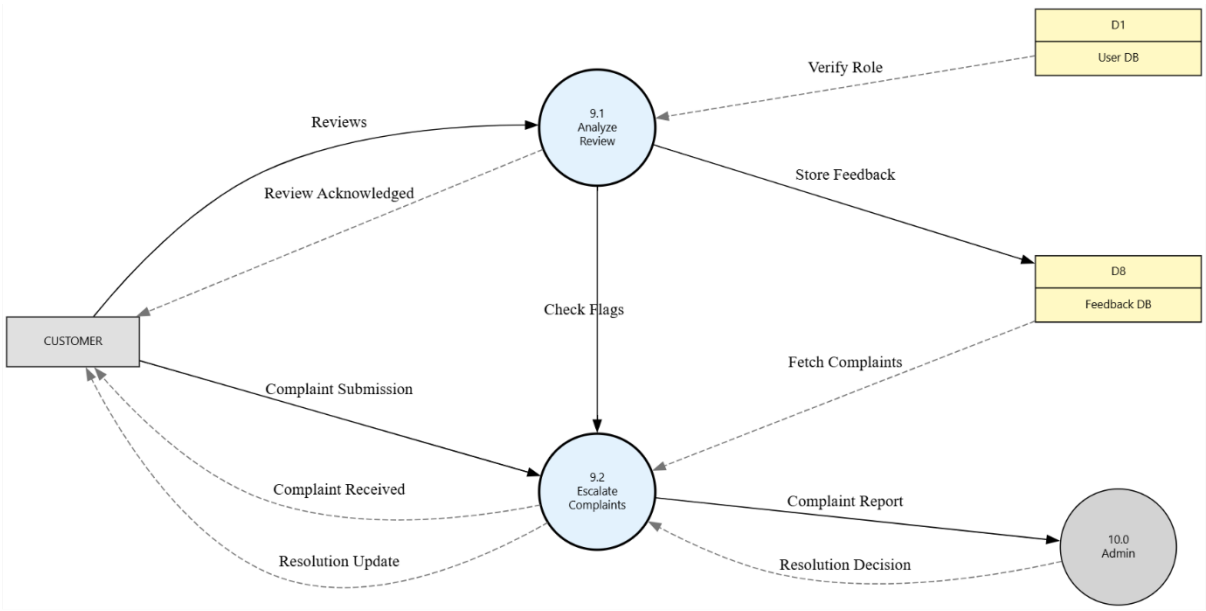
Process 7: Delivery Management:



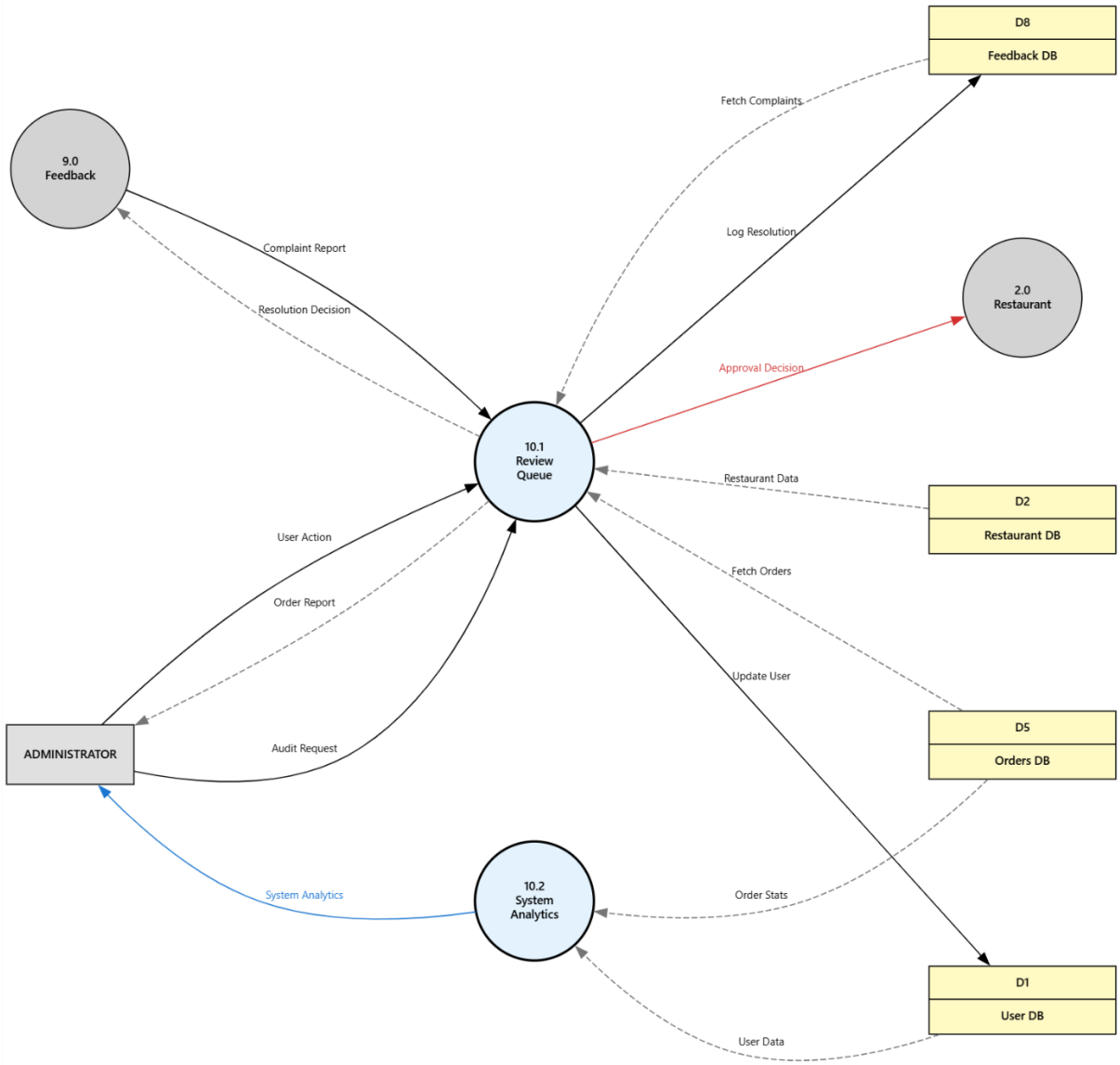
Process 8: Order Tracking:



Process 9: Feedback Management:



Process 10: Admin Management:



Explanation:

Shows the detailed breakdown of major system processes in the Cravit Smart Food Ordering System.

- Includes main processes like Order, Payment, Delivery, Tracking, Feedback, and Admin Management.
- Each process is divided into smaller sub-processes to explain internal working clearly.
- The system interacts with external entities such as Customer, Restaurant, Delivery Person, Map/GPS Service, Payment Gateway, and Administrator.
- Different data stores like User DB, Orders DB, Cart DB, Payments DB, Delivery DB, and Feedback DB are used to store and retrieve data.
- It explains the complete flow of data from order placement to delivery, tracking, feedback, and system administration.

2. Data Dictionary

DD Symbols :

Symbol	Meaning
+	Composition (combine data items)
[ ]	Selection (choose one option)
( )	Optional data
{ }	Iteration (repeated items)
=	Equivalence
** **	Comment

1.Primitive Data Items:

Data Name	Definition
userID	number
restaurantID	number
orderID	number
cartID	number
paymentID	number
deliveryID	number
feedbackID	number
txnID	number
latitude	decimal
longitude	decimal
timestamp	date + time
date	year + month + day

time	hour + minute + second
pincode	number
email	text
phone	number
password	text
hashed-password	hash(password)

2. User Account Data:

Data Name	Definition
User	userID + name + email + phone + hashed-password + role + address
name	firstName + lastName
address	street + city + state + pincode
role	[Customer, RestaurantOwner, DeliveryPartner, Administrator]
new-user-rec	name + email + phone + password + role
reg-confirmation	userID + role + timestamp
verification	email + verification-method
verification-method	[OTP, EmailLink]
auth-success	userID + session-token
auth-failed	email + timestamp
session-token	token
fetch-role	userID
role-confirmed	role
update-log	userID + timestamp
update-data	name + phone + address
reset-confirmation	email + timestamp

3. Restaurant Data:

Data Name	Definition
Restaurant	restaurantID + restaurantName + restaurantAddress + cuisine-type + status
restaurantAddress	street + city + state + pincode
cuisine-type	{cuisine}*
data-for-review	restaurant-registration-data
save-draft	restaurant-registration-data
check-duplicate	restaurantName + address
sync-availability	restaurantID + current-status
current-status	[Open, Closed]
fetch-restaurants	{restaurantID}*
restaurant-registration-data	restaurantName + restaurantAddress + cuisine-type

4. Menu Data:

Data Name	Definition
MenuItem	itemID + itemName + category + price + availability
menu-record	restaurantID + {MenuItem}*
save-menu-items	{MenuItem}*
save-category	categoryName
delete-item	itemID
remove-item	itemID
update-menu	restaurantID + {MenuItem}*
save-item-status	itemID + availability

fetch-menu	restaurantID
availability	[Available, NotAvailable]
menu-updates	itemID + price + availability

5. Search Data:

Data Name	Definition
SearchQuery	keyword + (cuisine-filter)
cuisine-filter	cuisine-type
validate-query	keyword
no-results	keyword
log-search	keyword + timestamp
filter-criteria	[Price, Rating, Distance]
sort-by	[PriceSort, RatingSort, DistanceSort]
RestList	{restaurantID}*

6. Cart Data:

Data Name	Definition
Cart	cartID + customerID + {CartItem}*
CartItem	itemID + quantity + price
fetch-cart	cartID
cart-record	cartID + {CartItem}*

7. Order Data:

Data Name	Definition
Order	orderID + customerID + restaurantID + {OrderItem}* + delivery-address + orderStatus + orderAmount
OrderItem	itemID + quantity + price
create-order	customerID + {OrderItem}*
clean-data	{OrderItem}*
invalid-order	error-code
id-generated	orderID
order-acceptance	restaurantID + orderID
new-order-alert	orderID
order-confirmed	orderID + timestamp
order-record	orderID + orderStatus
order-details	orderID + {OrderItem}*
order-confirmation	orderID + status
delivery-address	street + city + state + pincode

8. Payment Data:

Data Name	Definition
Payment	paymentID + orderID + paymentMode + paymentStatus + paymentAmount
payment-selection	paymentMode
payment-confirmation	paymentID + paymentStatus
trigger-pay	orderID + paymentAmount

transaction-request	txnID + orderID + paymentAmount
raw-status	gatewayResponse
final-success	txnID + Success
payment-failed	txnID + Failed
retry-request	txnID
cod-selection	orderID
cod-confirmed	orderID
e-receipt	paymentID + orderID + paymentAmount
settlements	restaurantID + paymentAmount
log-txn	txnID + timestamp
success-fail	[Success, Fail]

9. Delivery Data:

Data Name	Definition
Delivery	deliveryID + orderID + deliveryPartnerID + deliveryStatus + currentLocation
delivery-record	deliveryID + orderID
location-update	latitude + longitude
delivery-assignments	orderID + deliveryPartnerID
assign-task	orderID + deliveryPartnerID
partner-final	deliveryPartnerID
pickup-det	orderID + restaurantID
live-status	deliveryStatus + location-update

delivery-complete	deliveryID + timestamp
mark-delivered	orderID + Delivered
log-delivery	deliveryID + timestamp
delivery-confirmed	deliveryID + Delivered

10. Tracking Data:

Data Name	Definition
tracking-updates	orderID + location-updates
coordinate-request	orderID
real-time-location-data	latitude + longitude
location-updates	latitude + longitude + timestamp
gps-coordinates	latitude + longitude
navigation-route-data	route-data
route-data	{gps-coordinates}*
sync-gps	gps-coordinates
status-feed	orderStatus
fetch-order-status	orderID
merged-info	orderStatus + location-updates
live-tracking	merged-info
tracking-update	merged-info
push-notification	notification-data
Notifications	{notification-data}*

11. Feedback Data:

Data Name	Definition
Feedback	feedbackID + orderID + customerID + rating + review + complaint
ratings-complaints	rating-data + complaint-data
rating-data	rating + review
complaint-data	complaint + issue-type
store-feedback	Feedback
review-acknowledged	feedbackID
check-flags	complaint-data
complaint-received	complaint-data
fetch-complaints	{complaint-data}*
resolution-update	resolutionStatus
issue-type	[LateDelivery, FoodQuality, PaymentIssue]
reviews	{rating-data}*

12. Admin Monitoring Data:

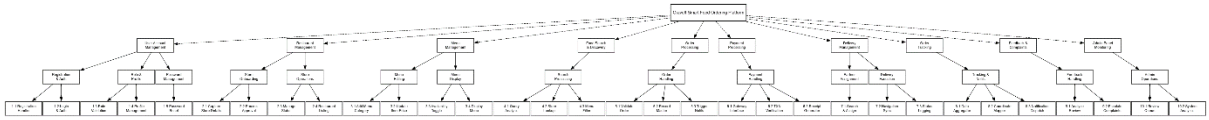
Data Name	Definition
approval-requests	restaurant-registration-data
system-queries	query-type
query-type	[UserQuery, OrderQuery, RestaurantQuery]
report-data	monitoring-data

reports	{report-data}*
monitoring-data	totalUsers + totalOrders + totalRestaurants + revenue
order-monitoring	orderID + orderStatus

13. Data Store Definitions:

Data Store	Definition
D1 User DB	{User}*
D2 Restaurant DB	{Restaurant}*
D3 Menu DB	{menu-record}*
D4 Cart DB	{cart-record}*
D5 Orders DB	{Order}*
D6 Payments DB	{Payment}*
D7 Delivery DB	{Delivery}*
D8 Feedback DB	{Feedback}*

3. Module Structure Chart



Drive link of above diagram:

https://drive.google.com/file/d/1dLKa_Qxqxygz0kLuN2mDcaJrRBrujSEc/view?usp=drive_link

Explanation:

- ROOT module**:** CraveIt Smart Food Ordering Platform.
- Main modules (10): User Account, Restaurant, Menu, Food Search, Order Processing, Payment, Delivery, Order Tracking, Feedback, Admin Panel.
- Sub-modules: Each main module splits into specific tasks (e.g., Registration, Role/Profile, Password, Store Onboarding, Menu Editing, Order Validation, Payment, Delivery Assignment, Tracking, Feedback Handling, Admin Operations).
- Functional components: Detailed processes under sub-modules implement core operations.
- Design: Hierarchical, modular, scalable, with clear separation of responsibilities.

4. Traceability Matrix

Requirement ID	Title	Description	Module	Priority	Risk
FR-UM-01	User Registration	System allows customers, restaurants, and delivery partners to create accounts.	User & Account Management	1	Critical
FR-UM-02	User Login	System authenticates users and grants role-based access.	User & Account Management	1	Critical
FR-UM-03	Password Reset	Users can reset forgotten passwords through verification.	User & Account Management	2	Medium
FR-UM-04	Role-Based Access	System restricts functionality according to user roles.	User & Account Management	1	Critical
FR-UM-05	Profile Management	Users can update personal and contact details.	User & Account Management	2	Low
FR-RM-01	Restaurant Registration	Restaurant owners can register restaurants on the platform.	Restaurant Management	1	High
FR-RM-02	Restaurant Approval	Admin approves or rejects restaurant registrations.	Restaurant Management	1	High
FR-RM-03	Restaurant Listing	Available restaurants are displayed to customers.	Restaurant Management	1	High
FR-RM-04	Restaurant Availability	Restaurants can mark themselves available or unavailable.	Restaurant Management	2	Medium

FR-MM-01	Menu Item Management	Restaurants add, update, or remove menu items.	Menu Management	1	High
FR-MM-02	Menu Pricing Update	Restaurants update item pricing and availability.	Menu Management	1	High
FR-MM-03	Menu Display	Customers can view menu details and prices.	Menu Management	1	High
FR-FD-01	Restaurant Search	Customers search restaurants by name or cuisine.	Food Search & Discovery	2	Medium
FR-FD-02	Food Search	Customers search food items across restaurants.	Food Search & Discovery	2	Medium
FR-FD-03	Filtering & Sorting	Restaurants can be filtered or sorted.	Food Search & Discovery	3	Low
FR-OP-01	Add to Cart	Customers add items to shopping cart.	Cart & Order Placement	1	Critical
FR-OP-02	Modify Cart	Customers modify or remove cart items.	Cart & Order Placement	1	Critical
FR-OP-03	Order Placement	Customers place immediate or scheduled orders using cart items.	Cart & Order Placement	1	Critical
FR-OP-04	Address Selection	Customers select or add delivery addresses.	Cart & Order Placement	1	High
FR-OP-05	Order Confirmation	System confirms successful orders.	Cart & Order Placement	1	High

FR-PM-01	Payment Mode Selection	Customer selects preferred payment method (online payment or cash on delivery) before confirming the order.	Payment Management	1	High
FR-PM-02	Payment Processing	System processes online payments through secure gateways and supports cash on delivery orders.	Payment Management	1	Critical
FR-PM-03	Payment Confirmation	System confirms successful payment for online transactions and confirms order placement for cash on delivery.	Payment Management	1	Critical
FR-PM-04	Payment Failure Handling	Users are notified when online payment fails and are allowed to retry the transaction.	Payment Management	2	Medium
FR-OR-01	Order Notification	Restaurants receive order notifications.	Order Processing (Restaurant)	1	Critical
FR-OR-02	Accept/Reject Order	Restaurants accept or reject orders.	Order Processing (Restaurant)	1	Critical
FR-OR-03	Preparation Update	Restaurants update preparation status.	Order Processing (Restaurant)	2	Medium
FR-DM-01	Delivery Assignment	System assigns delivery partners to orders.	Delivery Management	1	Critical
FR-DM-02	Delivery Partner Login	Delivery partners access assigned orders.	Delivery Management	1	High

FR-DM-03	Delivery Acceptance	Delivery partner accepts assigned order.	Delivery Management	1	High
FR-DM-04	Delivery Status Update	Delivery progress is updated by partner.	Delivery Management	1	High
FR-DM-05	Delivery Completion	Delivery partner marks order delivered.	Delivery Management	1	High
FR-TN-01	Order Tracking	Customers track order progress.	Tracking & Notifications	1	High
FR-TN-02	GPS Tracking	Customers view delivery partner location.	Tracking & Notifications	2	Medium
FR-TN-03	Notifications	System sends order update notifications.	Tracking & Notifications	2	Medium
FR-FC-01	Rating Submission	Customers submit ratings and reviews.	Feedback & Complaint	3	Low
FR-FC-02	Complaint Submission	Customers submit complaints.	Feedback & Complaint	2	Medium
FR-FC-03	Complaint Handling	Admin reviews and resolves complaints.	Feedback & Complaint	2	Medium
FR-AM-01	User Management	Admin manages users and partners.	Admin Management	1	Critical
FR-AM-02	Restaurant Management	Admin monitors restaurants.	Admin Management	1	High

FR-AM-03	Order Monitoring	Admin monitors all orders.	Admin Management	1	High
FR-AM-04	Analytics Dashboard	Admin views performance statistics.	Admin Management	2	Medium
FR-HR-01	Order History Storage	System stores customer order history.	Order History & Reports	2	Low
FR-HR-02	Order Reordering	Customers reorder past orders.	Order History & Reports	3	Low